

Lab07_Barbosa

Danilo Barbosa

Laboratório 3: SQLite II

1. Baixe o arquivo `uwmadison.sqlite3`. Conecte-se a ele usando o pacote `RSQLite`, armazenando a conexão em uma variável `conn`.

```
library(RSQLite)

conn <- dbConnect(SQLite(), "uwmadison.sqlite3")
```

2. Quem são e quantos são os professores que lecionaram disciplinas cujo tópico era estatística (`subjects.abbreviation='STAT'`);

```
dbGetQuery(conn, "SELECT
    i.id,
    i.name
FROM instructors i
JOIN teachings t
    ON i.id = t.instructor_id
JOIN sections sec
    ON t.section_uuid = sec.uuid
JOIN subject_memberships sm
    ON sec.course_offering_uuid = sm.course_offering_uuid
JOIN subjects sub
    ON sm.subject_code = sub.code
WHERE sub.abbreviation = 'STAT'
GROUP BY i.id, i.name")
```

	id	name
1	1079018	JUN SHAO
2	1114818	MICHAEL FERRIS
3	1126199	JOSEP GINEBRA

4	129415	ANATOLE BECK
5	138086	CHRISTINE SORKNESS
6	1426304	ZHANG CHUNMING
7	143932	ISMOR FISCHER
8	1505727	NATALIA DE LEON GATTI
9	1505756	GUILHERME ROSA
10	1515839	PHILIP WOOD
11	1634418	JONATHON PETERSON
12	207204	KYUNGMAH KIM
13	232828	NELLIE LAUGHLIN
14	245914	MARIAN R FISHER
15	2600141	RICHARD A BRUALDI
16	2600233	DIETRICH UHLENBROCK
17	2600488	GRACE WAHBA
18	2600689	NORMAN DRAPER
19	2600893	ARNOLD MILLER
20	2600973	STEFFEN LEMPP
21	2601039	THOMAS G. KURTZ
22	2601051	STEPHEN WAINGER
23	2601091	ROBERT WARDROP
24	2601093	PAUL M TERWILLIGER
25	2601512	ROBERT R MEYER
26	2601596	DAVID DEMETS
27	2601613	RICHARD A. JOHNSON
28	2601621	MURRAY CLAYTON
29	2601633	JAMES D KUELBS
30	2601674	MARI PALTA
31	2601916	MICHAEL NEWTON
32	2601926	RICHARD J. CHAPPELL
33	2602032	KAM-WAH TSUI
34	2602042	ERIK NORDHEIM
35	2602050	DOUGLAS M. BATES
36	2602055	BRIAN YANDELL
37	2602108	WEI-YIN LOH
38	2602109	DAVID GRIFFEATH
39	270258	MICHAEL HOGAN
40	2824494	ELOISA D CHAVAS
41	3010828	YU MENGANG
42	3011936	JASON P FINE
43	3077702	MING YUAN
44	3095399	JUNGWON MUN
45	3103459	LANCINE KONATE
46	3107260	ZHENGXIAO WU

47	3111842	ADIN-CRISTIAN ANDREI
48	3117835	SUKHENDU MEHROTRA
49	3122658	STEPHEN AARON STANHOPE
50	3127329	KEN ONO
51	3127330	SHI JIN
52	321374	JUN ZHU
53	3230527	TONGHAI YANG
54	3253489	SCOTT JOSEPH HETZEL
55	333334	MARY LINDSTROM
56	3338237	QIAN ZHIGUANG
57	3339239	ZHIGUO XIAO
58	3343005	XIAODAN WEI
59	3343055	RUIFENG XU
60	3351080	TING-LI LIN
61	3374880	TIMO SEPPALAINEN
62	3376532	CHELSEY GREEN
63	3382451	STEPHEN WRIGHT
64	3404298	MOO K CHUNG
65	3404299	KJELL DOKSUM
66	340544	CHRISTINA M. KENDZIORSKI
67	3437432	LEV BORISOV
68	3443667	LEI XU
69	3456935	HAODA FU
70	3457484	YANBING ZHENG
71	3460189	FANG FANG
72	3467530	MATTHEW BALLARD
73	3483534	HUI WANG
74	3484236	BO HUANG
75	3523865	NICHOLAS HENDERSON
76	3566378	ERICA LEE DEADMAN
77	3573775	ALEXANDER KISELEV
78	3573852	SEUNGBONG HAN
79	3575433	MIN JUNG LEE
80	3613299	CHIEN-WEI CHEN
81	361804	FREDERICK BOEHM
82	3620114	JUN YIN
83	3620552	JUNHEE HAN
84	3620612	XINWEI DENG
85	3622809	SIJIAN WANG
86	3623432	HUAIBAO FENG
87	3626007	ZHUANG WU
88	3627934	JESSE THOMAS HOLZER
89	3629214	CHIA-CHIEH LIN

90	3661782	ALBRECHT KLEMM
91	3668575	XUN ZHAO
92	3674168	ANDREJ ZLATOS
93	3727371	RYAN ZEA
94	3773167	DEYUAN JIANG
95	3775809	AKICHIKA OZEKI
96	3777992	KAZUHIKO SHINKI
97	3780010	ROBERT WAYNE GREEN
98	3780817	WEI ZHENG
99	3781422	TAO YU
100	3781764	JUN ZHANG
101	3783301	FANGFANG WANG
102	3783649	JINGCI MENG
103	3791787	YUAN JIANG
104	3791802	XIWEN MA
105	3800708	YALI WANG
106	3801422	SUNDUZ KELES
107	3806328	JOHN KANE
108	3855025	CHRISTOPHER WAGNER
109	3882760	NORBERT BINKIEWICZ
110	3884290	SRIKANTHMADHAVAN ARAVAMUTHAN
111	393380	DONALD PORTER
112	394810	MICHELLE HARRIS
113	3955531	SONA Z SWANSON
114	3955546	CECILE ANE
115	3956063	JASON RICHARD SWANSON
116	3959251	VARSHA KULKARNI
117	3961448	YU QIU LIU
118	3965653	MIN NIU
119	3973318	JINGJIANG PENG
120	3973813	CHENXI LI
121	3974100	QI TANG
122	3976099	CHANHAN HSU
123	3978606	KEVIN HASEGAWA ENG
124	3980675	LILI LAN
125	3981051	JUN LI
126	3981781	SANGBUM CHOI
127	3987897	RUIFANG SONG
128	4108525	ZHENGJUN ZHANG
129	4128750	RUI TANG
130	4133622	MING XIE
131	4138542	ZUOFENG SHANG
132	4140397	JEE YEON KIM

133	4140469	BEN ADAM HAALAND
134	4144479	YUJIN CHUNG
135	4145889	XIPEI YANG
136	4146030	SHENG WANG
137	4149787	PERLA REYES
138	4149843	HEATHER MARIE BRAZEAU
139	4155990	BIN DAI
140	4161984	QUOC TRAN
141	417071	THOMAS COOK
142	4172260	YANNAN QIU
143	4194670	XIUFENG SHAO
144	4195000	MARIA KAMENETSKY
145	4199106	KRISTEN CYFFKA
146	4261114	KEEGAN KORTHAUER
147	4264265	DEREK BEAN
148	4271924	SHUANG HUANG
149	4272783	GUANHUA CHEN
150	4277002	LIE XIONG
151	4278664	YOUNGDEOK HWANG
152	4280220	MENG SONG
153	4285354	YANG ZHAO
154	4285528	XIN LI
155	4285538	YI CHAI
156	4285558	SHENG ZHANG
157	4286103	JIWEI ZHAO
158	4286138	XU HE
159	4286679	XINXIN YU
160	4292460	JIE WEI
161	4308515	DAVID ANDERSON
162	4312759	COLE COOK
163	4392906	MINJING TAO
164	4426932	YAJUAN SI
165	4430246	XINYU SONG
166	4432994	QUEFENG LI
167	4433317	JIALE XU
168	4434130	WAI TONG FAN
169	4435446	LAM HO
170	4435790	HAO ZHENG
171	4436002	GUANNAN SUN
172	4436066	BIN ZHU
173	4436368	JIE ZHANG
174	4437355	FAN YANG
175	4460098	QIONG ZHANG

176	4491860	KATHERINE GOODE
177	4538519	BENEDEK VALKO
178	4538540	STEPHEN BERG
179	4541610	KARL BROMAN
180	4541633	JUNHO LEE
181	4576059	YAoyao XU
182	4578330	LILUN DU
183	4582107	QIURONG CUI
184	4586128	YI LIU
185	4586558	CHEN CHENG
186	4588103	XIN ZHANG
187	4588354	SHUYUN YE
188	4588525	ZHIGENG GENG
189	4590481	WENWEN ZHANG
190	4590559	XU XU
191	4590715	JIAJIE CHEN
192	4590738	TAERI UHM
193	4591139	XIAO GUO
194	4593965	SOKOL VAKO
195	4597046	CHENLIANG XU
196	4598772	SHIZHEN WANG
197	4601948	FLORIAN BERTRAND
198	4603165	JIAN WU
199	4610245	JOSEPH DEUTSCH
200	4613728	JENNIFER NGUYEN
201	4632211	BENJAMIN RECHT
202	4633355	QI JIANG
203	4636145	JENNIFER BIRSTLER
204	4636315	JARED BROWN
205	4641200	VICTOR LUO
206	4646729	ANRU ZHANG
207	4680082	ALEXANDER FISH
208	4695492	YAZHEN WANG
209	4699427	SHUAI CHEN
210	4712098	CHAOYANG YU
211	4721578	CLAUDIA SOLIS LEMUS
212	4721620	YAN CHEN
213	4724058	CHAOQUN MEI
214	4725072	HAOYANG FAN
215	4726666	XINJIE HE
216	4726678	LU MAO
217	4726703	XUEYAO CHEN
218	4728844	DUY NGUYEN

219	4729481	XIAOPING FENG
220	4731731	JU HEE CHO
221	4733192	SEHO PARK
222	4733966	YUAN WANG
223	4734274	JILI WANG
224	4734607	HYUNSEUNG KANG
225	4744350	KEVIN PACKARD
226	4752238	SAMUEL STECHMANN
227	4753018	BRET HANLON
228	4761200	GREGORIO MORENO-FLORES
229	4787911	ABIGAIL BENZINE
230	4788841	VICTORIA MANSFIELD
231	4845798	LIAM JOHNSTON
232	4879129	TIMOTHY IDOWU
233	4880968	JOHN DAVIS
234	4884161	GUN WOONG PARK
235	4885477	SHIXUE LIU
236	4886028	GUILHERME VIEIRA NUNES LUDWIG
237	4886987	JIANCHANG HU
238	4890283	JEEA CHOI
239	4895244	YAOGUO XIE
240	4896172	TRAM TA
241	4896511	JOHN WILTSHIRE-GORDON
242	4896939	TIEN VO
243	4897725	DONGGYU KIM
244	4898014	DONG XIA
245	4902482	JINGLAN LI
246	4903489	CHENSHENG KUANG
247	4904255	THU LE
248	4905505	XIAO NIE
249	4906458	YUANZHI LI
250	4929500	KARL ROHE
251	4947463	MITCHELL PAUKNER
252	495087	GARY HOWARD SCHROEDER
253	4952769	ANNE BRUCKNER
254	4953246	SONG WANG
255	4957688	GENG LI
256	4957895	ROBERT HARRON
257	4957916	TIMOTHY IDOWU
258	4958796	ALAN HUANG
259	4963378	AUGUST JENSEN
260	5003565	ANDREW LESLIE
261	5010171	HAN CHEN

262	5010560	LU YANG
263	5011312	BROOK LUERS
264	5012817	KAILEI CHEN
265	5015113	LONG PHAN
266	5015194	CUICUI QI
267	5017571	SHENGJI JIA
268	5017655	GINA BENNINGER
269	5017774	LUWAN ZHANG
270	5018463	CHEN JING
271	5020061	RUOSI GUO
272	5023414	THEVAASIINEN CHANDERENG
273	5068296	SEAN KENT
274	5099221	ZIFENG ZHAO
275	5110390	SEBASTIEN ROCH
276	5127444	GREGORY SHINAULT
277	5144884	BINGYING XIE
278	5161294	WESLEY CHANG
279	5162339	ALYSSA DIGILIO
280	5167906	MARIA ROJO
281	5167973	YUAN LI
282	5167990	HUILIN HU
283	5168069	HUIKUN ZHANG
284	5170492	YING ZHANG
285	5170801	XIAOMAO LI
286	5170885	SHULEI WANG
287	5170924	XIRAN WANG
288	5176022	CUIZE HAN
289	5176905	SOHEIL SADEGHI
290	5177493	YOUNG MIN PARK
291	5186047	BIN ZHANG
292	5189535	EDWARD ERKER
293	5192960	LUXI CAO
294	5204089	FAN GAO
295	5214164	GARVESH RASKUTTI
296	5214858	YUQING XU
297	5258581	XIAOWU DAI
298	5302154	MANJUSHA KANCHARLA
299	5305675	DEREK NORTON
300	5317373	KYLE HEBERT
301	5320625	GONZALO CONTADOR
302	5321537	HYEBIN SONG
303	5322094	WENZHI CAO
304	5322566	XIUYU MA

305	5323628	SIJING LI
306	5323690	HAO ZHOU
307	5323691	YILIN ZHANG
308	5326059	YOURAN QI
309	5326263	PEIGEN ZHOU
310	5327139	YI LI
311	5329574	TONG LI
312	5332940	COLIN LONGHURST
313	5354849	SCOTT HOTTOVY
314	5410648	JIE SONG
315	5451706	ALBERTO DEL PIA
316	5452538	STEVEN SAM
317	5476019	KRISHNAKUMAR BALASUBRAMANIAN
318	5478075	KYLE HERBET
319	5488796	REBECCA POST
320	5498092	ANQI SHI
321	5498648	CRYSTAL CHEN
322	5514514	HAO CHEN
323	5514517	YOUNG LEE
324	5514561	MUHONG GAO
325	5514932	LIN QI
326	5515201	SHANG WU
327	5515948	JAMES ANDERSON
328	5516726	DUZHE WANG
329	5518378	YONGSU LEE
330	5520701	HAO TENG
331	5520811	SEULKEE YUN
332	5520995	TING YE
333	5521838	LAN LUO
334	5523214	NING FAN
335	5567363	TUN LEE NG
336	5591917	ZIJIAN NI
337	5592111	CHENGNING ZHANG
338	5620719	DONGHYUN LEE
339	5631398	PAUL SAVARIAPPAN
340	5632570	BO YANG
341	5644228	YONGJOON KIM
342	5656034	ALEXANDER COVINGTON
343	5664494	TZU HSIANG HUNG
344	5690182	RUI CHEN
345	5690247	BOWEN HU
346	5693114	LILI ZHENG
347	5693277	YIFAN MEI

348	5693629	ZHONGJIE YU
349	5694563	BEHZAD AALIPUR HAFSHEJANI
350	5696436	YONGFENG WU
351	5697552	KUNLING HUANG
352	5700058	YUCHEN ZHOU
353	5708167	MICHAEL KUTZLER
354	5718599	QING LI
355	5840930	MICHAEL LIOU
356	5846795	DANIELE CAPPELLETTI
357	5852265	BEHZAD AALIPUR
358	5862942	GINA OH
359	5883206	CHAN PARK
360	5887049	CLAIRE BOBST
361	5887858	RONGJUN ZHU
362	5898424	RUNGANG HAN
363	5903922	YUCHANG WU
364	5905014	SHAN LU
365	5906523	BI CHENG WU
366	5906594	SHENG WANG
367	5907599	YUNONG LIN
368	6029366	MIHAELA IFRIM
369	6043299	DEBRAJ DAS
370	608997	JOHN GILLETT
371	640407	NICHOLAS STEPHEN KEULER
372	751075	BRET LARGET
373	783769	MICHAEL GEORGE ILTIS
374	789363	REBECCA KOSCIK
375	801445	IAN BRANSTAD RILEY
376	893827	RONALD GANGNON
377	967335	PAUL RATHOUZ
378	982957	SHANE HUBLER
379	989414	MICHAEL RENE KOSOROK

```

dbGetQuery(conn, "SELECT COUNT(DISTINCT i.id) AS total_professores
FROM instructors i
JOIN teachings t ON i.id = t.instructor_id
JOIN sections sec ON t.section_uuid = sec.uuid
JOIN subject_memberships sm ON sec.course_offering_uuid = sm.course_offering_uuid
JOIN subjects sub ON sm.subject_code = sub.code
WHERE sub.abbreviation = 'STAT'")

```

total_professores

3. O GPA americano é definido numa escala de 0 a 4, em que A = 4, AB = 3.5, B = 3, BC = 2.5, C = 2, D = 1 e F = 0. Determinando a nota média de cada oferecimento pela ponderação da quantidade de alunos em cada extrato com os valores numéricos de cada conceito, indique (no que se referente a disciplinas no assunto de estatística):

- Quem é o professor mais difícil?

```
dbGetQuery(conn, "SELECT
  i.name AS professor,
  ROUND(
    CAST(
      SUM(gd.a_count * 4.0 + gd.ab_count * 3.5 + gd.b_count * 3.0 +
        gd.bc_count * 2.5 + gd.c_count * 2.0 + gd.d_count * 1.0) AS FLOAT
    ) / NULLIF(SUM(gd.a_count + gd.ab_count + gd.b_count +
      gd.bc_count + gd.c_count + gd.d_count + gd.f_count), 0),
    2
  ) AS gpa_medio
FROM instructors i
JOIN teachings t ON i.id = t.instructor_id
JOIN sections sec ON t.section_uuid = sec.uuid
JOIN grade_distributions gd ON sec.course_offering_uuid = gd.course_offering_uuid AND sec.number = gd.offering_number
JOIN subject_memberships sm ON sec.course_offering_uuid = sm.course_offering_uuid
JOIN subjects sub ON sm.subject_code = sub.code
WHERE sub.abbreviation = 'STAT'
GROUP BY i.id, i.name
ORDER BY gpa_medio ASC
LIMIT 1")
```

	professor	gpa_medio
1	CHRISTINE SORKNESS	NA

- Quem é o professor mais fácil?

```
dbGetQuery(conn, "SELECT
  i.name AS professor,
  ROUND(
    CAST(
      SUM(gd.a_count * 4.0 + gd.ab_count * 3.5 + gd.b_count * 3.0 +
        gd.bc_count * 2.5 + gd.c_count * 2.0 + gd.d_count * 1.0) AS FLOAT
    ) / NULLIF(SUM(gd.a_count + gd.ab_count + gd.b_count +
```

```

                gd.bc_count + gd.c_count + gd.d_count + gd.f_count), 0),
            2
        ) AS gpa_medio
FROM instructors i
JOIN teachings t ON i.id = t.instructor_id
JOIN sections sec ON t.section_uuid = sec.uuid
JOIN grade_distributions gd ON sec.course_offering_uuid = gd.course_offering_uuid AND sec.num_students = gd.num_students
JOIN subject_memberships sm ON sec.course_offering_uuid = sm.course_offering_uuid
JOIN subjects sub ON sm.subject_code = sub.code
WHERE sub.abbreviation = 'STAT'
GROUP BY i.id, i.name
ORDER BY gpa_medio DESC
LIMIT 1")

```

	professor	gpa_medio
1	SRIKANTHMADHAVAN ARAVAMUTHAN	4

- Qual é a disciplina mais difícil?

```

dbGetQuery(conn, "SELECT
    c.name AS disciplina,
    ROUND(
        CAST(
            SUM(gd.a_count * 4.0 + gd.ab_count * 3.5 + gd.b_count * 3.0 +
                gd.bc_count * 2.5 + gd.c_count * 2.0 + gd.d_count * 1.0) AS FLOAT
        ) / NULLIF(SUM(gd.a_count + gd.ab_count + gd.b_count +
            gd.bc_count + gd.c_count + gd.d_count + gd.f_count), 0),
        2
    ) AS gpa_medio
FROM courses c
JOIN course_offerings co ON c.uuid = co.course_uuid
JOIN grade_distributions gd ON co.uuid = gd.course_offering_uuid
JOIN subject_memberships sm ON co.uuid = sm.course_offering_uuid
JOIN subjects sub ON sm.subject_code = sub.code
WHERE sub.abbreviation = 'STAT'
GROUP BY c.uuid, c.name
ORDER BY gpa_medio ASC
LIMIT 1")

```

	disciplina	gpa_medio
1	Directed Study	NA

- Qual é a disciplina mais fácil?

```
dbGetQuery(conn, "SELECT
  c.name AS disciplina,
  ROUND(
    CAST(
      SUM(gd.a_count * 4.0 + gd.ab_count * 3.5 + gd.b_count * 3.0 +
        gd.bc_count * 2.5 + gd.c_count * 2.0 + gd.d_count * 1.0) AS FLOAT
    ) / NULLIF(SUM(gd.a_count + gd.ab_count + gd.b_count +
      gd.bc_count + gd.c_count + gd.d_count + gd.f_count), 0),
    2
  ) AS gpa_medio
FROM courses c
JOIN course_offerings co ON c.uuid = co.course_uuid
JOIN grade_distributions gd ON co.uuid = gd.course_offering_uuid
JOIN subject_memberships sm ON co.uuid = sm.course_offering_uuid
JOIN subjects sub ON sm.subject_code = sub.code
WHERE sub.abbreviation = 'STAT'
GROUP BY c.uuid, c.name
ORDER BY gpa_medio DESC
LIMIT 1")
```

	disciplina	gpa_medio
1	Nonparametric Statistics and Machine Learning Methods	4

4. Desconecte do banco de dados.

```
dbDisconnect(conn)
```

```
Sys.setenv(TZ = "America/Sao_Paulo")
Sys.time()
```

```
[1] "2026-09-17 11:16:08 -03"
```